

Expt 8

Multirate Signal Processing

8.1 Implementation and Analysis of Decimation and Interpolation in Multirate Signal Processing Using MATLAB

Objectives:

1. To generate a discrete-time sinusoidal sequence for multirate signal processing.
2. To perform decimation of a discrete-time sequence by a factor of 2.
3. To perform Interpolation of a discrete-time sequence by a factor of 2.
4. To analyze and compare the original, decimated, and interpolated sequences.

Procedure

1. Open MATLAB and initialize the workspace using `clc`, `clear`, and `close all`.
2. Generate a discrete-time sinusoidal sequence with the desired frequency and sampling rate.
3. Perform decimation by selecting every Mth sample using the `downsample()` function.
4. Perform interpolation by inserting zeros and filtering using the `upsample()` function.
5. Display the original signal samples.
6. Display the decimated signal samples.
7. Display the interpolated signal samples.
8. Plot and compare the original, decimated, and interpolated signals.

Program (Matlab Code):

```
clc;
```

```
clear;
```

```
close all;
```

```
%% Original Sequence
```

```
n = 0:40;
```

```
x = sin(0.3*pi*n);
```

```
%% Decimation by 2
```

```
y = x(1:2:end);
```

```
nd = 0:length(y)-1;
```

```

%% Interpolation by 2

v = zeros(1, 2*length(x));

v(1:2:end) = x;

ni = 0:length(v)-1;


%% Figure 1: Original Sequence

figure;

stem(n, x, 'filled');

grid on;

title('Original Sequence x[n]');

xlabel('n');

ylabel('Amplitude');


%% Figure 2: Decimated Sequence

figure;

stem(nd, y, 'filled');

grid on;

title('Decimated Sequence by 2');

xlabel('n');

ylabel('Amplitude');


%% Figure 3: Interpolated Sequence

figure;

stem(ni, v, 'filled');

grid on;

title('Interpolated Sequence by 2');

xlabel('n');

ylabel('Amplitude');


%% Figure 4: Comparison

```

```

figure;

stem(n, x, 'b');

hold on;

stem(0:2:length(v)-2, v(1:2:end), 'r');

legend('Original x[n]', 'Interpolated Samples');

grid on;

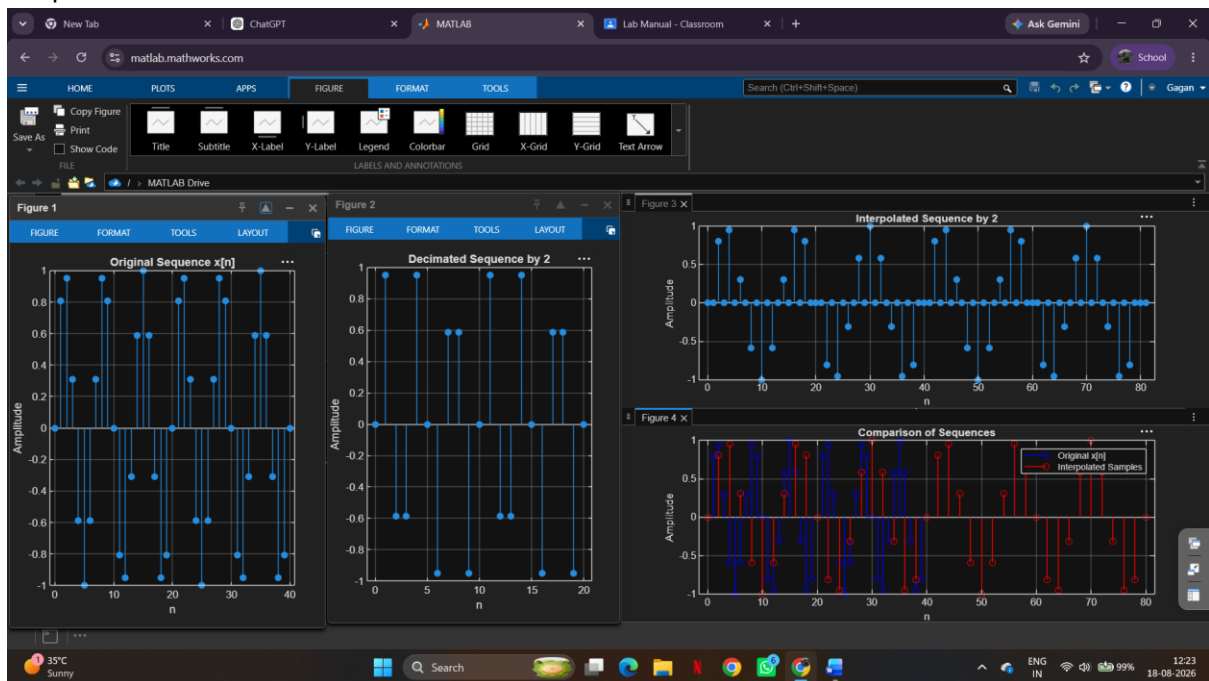
title('Comparison of Sequences');

xlabel('n');

ylabel('Amplitude');

```

Output:



Result :

1. A discrete-time sinusoidal sequence was successfully generated for multirate processing.
2. The discrete-time sequence was successfully decimated by a factor of 2, reducing the number of samples and the sampling rate.
3. The discrete-time sequence was successfully interpolated by a factor of 2, increasing the number of samples and the sampling rate.
- 4.. The effects of downsampling and upsampling were successfully analyzed by comparing the original, decimated, and interpolated signals.